

Football Analytics Platform

UBC MDS Capstone 2026 — Trilemma Foundation

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Trilemma Foundation & Our Team

Partner: Trilemma Foundation — non-profit advancing **evidence-based football development** worldwide, partnering with clubs, federations, and analysts using StatsBomb event data.

Name	Role
Quan Hoang	Data pipeline · dbt + Dagster
Li Pu	RQ1 · Player archetypes · Clustering
Rabin Duran	RQ2 · Consistency scoring · Explorer
Teem Kwong	RQ3 · Natural language chatbot

Our Dataset — StatsBomb Event Data

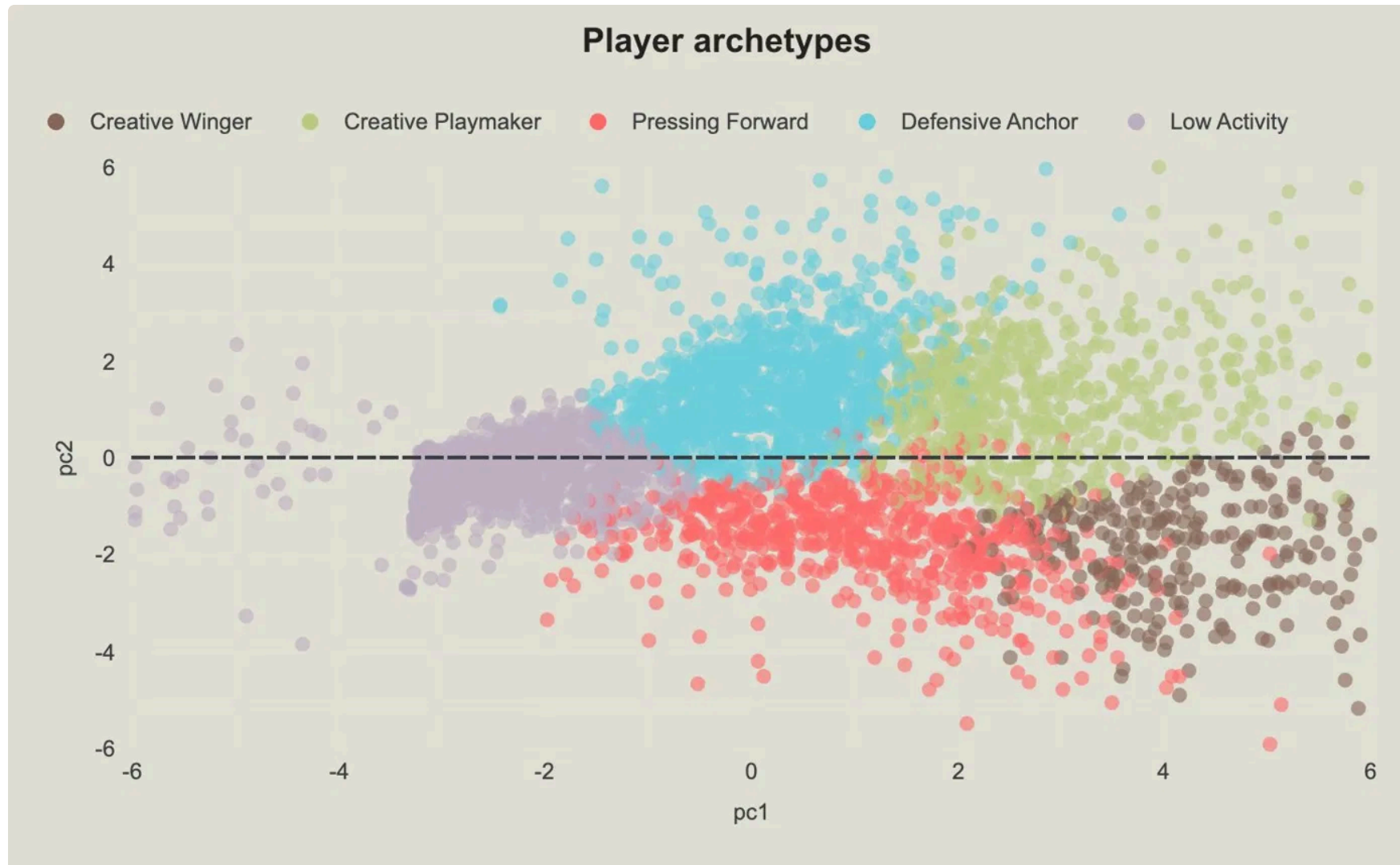
- **Source:** StatsBomb open event data — every on-ball action (pass, shot, pressure, dribble, tackle) tracked per match
- **Scale:** 15 competitions · ~3.5 M events · 5,800+ player-seasons across multiple seasons of La Liga, Premier League, Champions League, Bundesliga, and more
- **Aggregation for ML:** player-season level · filtered to **≥ 270 minutes** (~3 full matches) to ensure statistically reliable samples
- **Scope:** men's competitions only · goalkeepers excluded from archetype clustering

Problem & Research Questions

The gap: StatsBomb event data exists, but analysts need complex SQL to extract meaning.

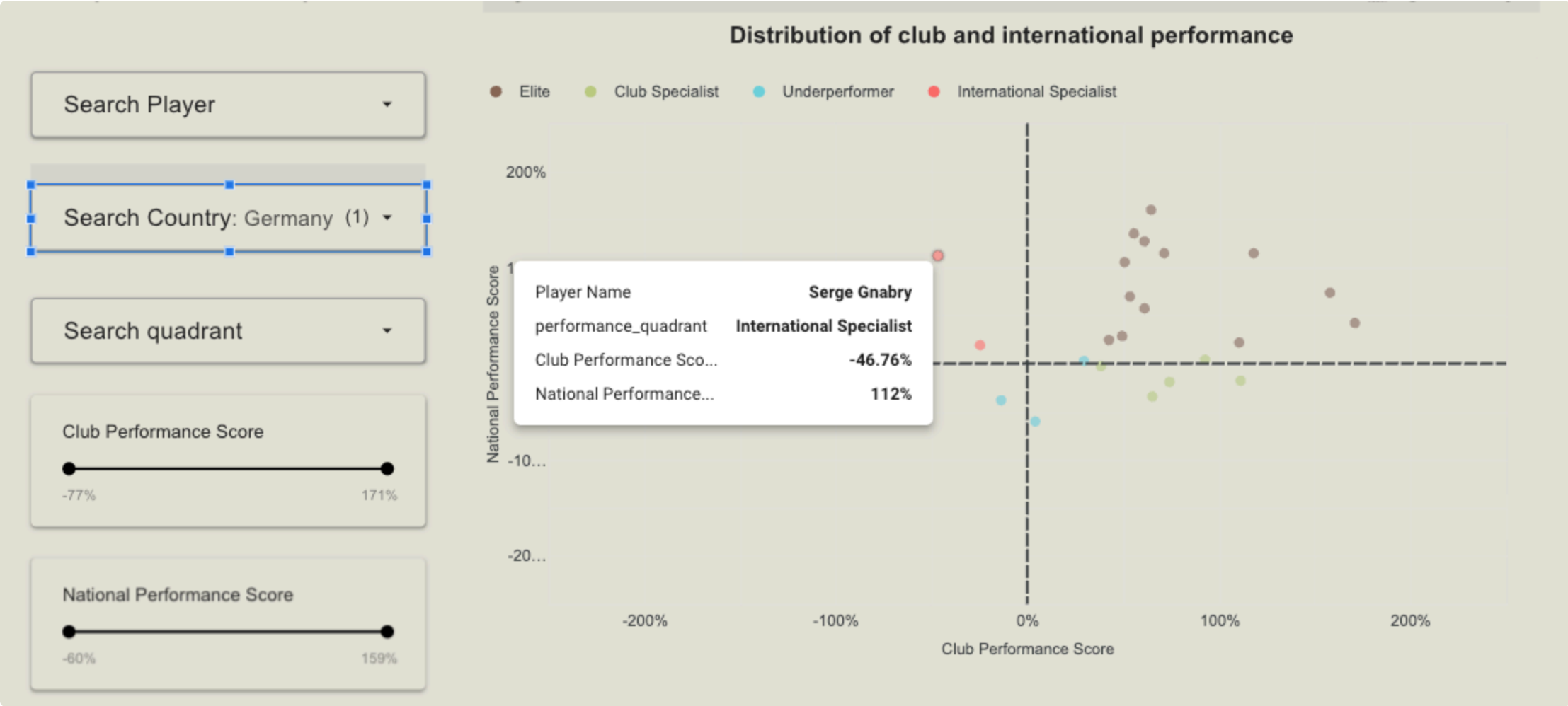
RQ	Question	Who	Product
RQ1	What tactical archetypes exist across competitions?	Li	Clustering + Looker dashboard
RQ2	Do players perform consistently club vs nationally?	Rabin	Consistency Explorer
RQ3	Can analysts query football data in plain English?	Teem	LLM chatbot
—	How does data get there automatically?	Quan	dbt + Dagster pipeline

RQ1: 5 Tactical Archetypes Found Across 15 Competitions



PCA scatter — 5 archetypes · 5,800+ player-seasons · 15 competitions · Li walks through the methodology

RQ2: Most Players Perform Differently Club vs Nationally



Performance score = weighted sum of Z-scores across 13 per-90 metrics · 493 players · 4 quadrants · Rabin covers the method

RQ3: Any Football Question, Answered in Plain English

The screenshot shows a web browser window with the URL `136.118.99.32`. The page title is "Elite European Football AI" with tabs for "Dashboard" and "Chatbot". The interface is powered by Groq & BigQuery.

SUGGESTED QUESTIONS

- Top 5 creative wingers sorted by xG
- Who are the Elite players in La Liga?
- What cluster does Messi belong to?
- Show me the top defensive anchors by pressures per 90
- Which players are Club Specialists?

Elite AI Assistant

1. Javier Alejandro Mascherano — pressures_per_90: 67.25
2. Pierre-Emile Højbjerg — pressures_per_90: 50.67
3. Ellyes Joris Skhiri — pressures_per_90: 42.33
4. Nicolò Barella — pressures_per_90: 39.27
5. André-Frank Zambo Anguissa — pressures_per_90: 39.25
6. Valon Behrami — pressures_per_90: 37.62
7. Lassana Coulibaly — pressures_per_90: 37.51
8. Ander C — Defensive pressure instances executed normalized per 90 minutes played. .0
9. Seydou
10. Nemanja Matić — pressures_per_90: 35.0

Executed SQL Query:

```
SELECT player_name, pressures_per_90 FROM `football-capstone-mds-496219.dbt_marts.mart_player_performance` WHERE cluster_label = 'Defensive Anchor' ORDER BY pressures_per_90 DESC LIMIT 10
```

Raw BigQuery Data:

```
[
  {
    "player_name": "Javier Alejandro Mascherano",
    "pressures_per_90": 67.25
  },
  ...
]
```

e.g., Which players match Rodri's tactical profile inside mart_player_clusters?

☒ Advanced Mode (Show SQL query & raw data)

Ask AI

Plain English → BigQuery SQL → formatted answer · under 10 sec · no SQL required · Teem demos live

Clustering Pipeline

RQ1: What player archetypes exist across European football?

Step	Detail
Input	<code>int_player_season_stats</code> from dbt
Filter	≥ 270 min · goalkeepers excluded
Features	13 per-90 metrics — see below
Model	Median impute · clip outliers · Scaler → PCA → KMeans k=5
Output	5 archetypes → <code>cluster_assignments</code> → Looker + chatbot

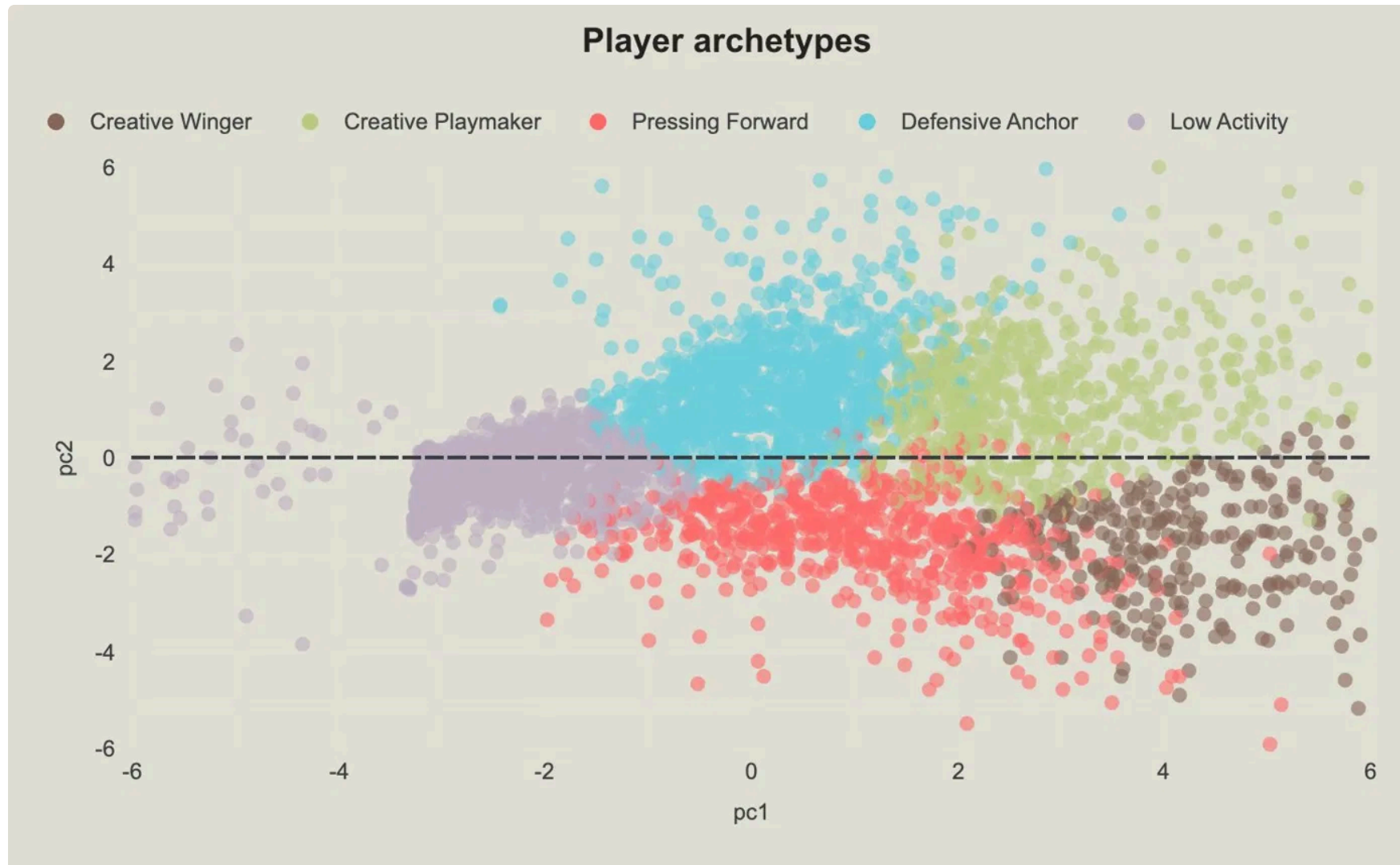
13 per-90 features: goals · xG · shots · passes · key passes · dribbles completed · pressures · tackles · interceptions · clearances · blocks · carries · aerial duels won

Cluster Results: 5 Archetypes

Per-90 metrics are prefixed `x_` in the dbt model (e.g. `x_goals`, `x_passes`, `x_pressures`).

Archetype	Defining <code>x_</code> features
Creative Winger	<code>x_xg</code> , <code>x_dribbles</code> , <code>x_passes</code> (att. third)
Creative Playmaker	<code>x_passes</code> , <code>x_carries</code> , <code>x_key_passes</code>
Pressing Forward	<code>x_pressures</code> , <code>x_shots</code> , <code>x_tackles</code>
Defensive Anchor	<code>x_interceptions</code> , <code>x_clearances</code> , <code>x_blocks</code>
Low Activity	Below-cohort average across most features

Cluster Scatter — What the Plot Shows



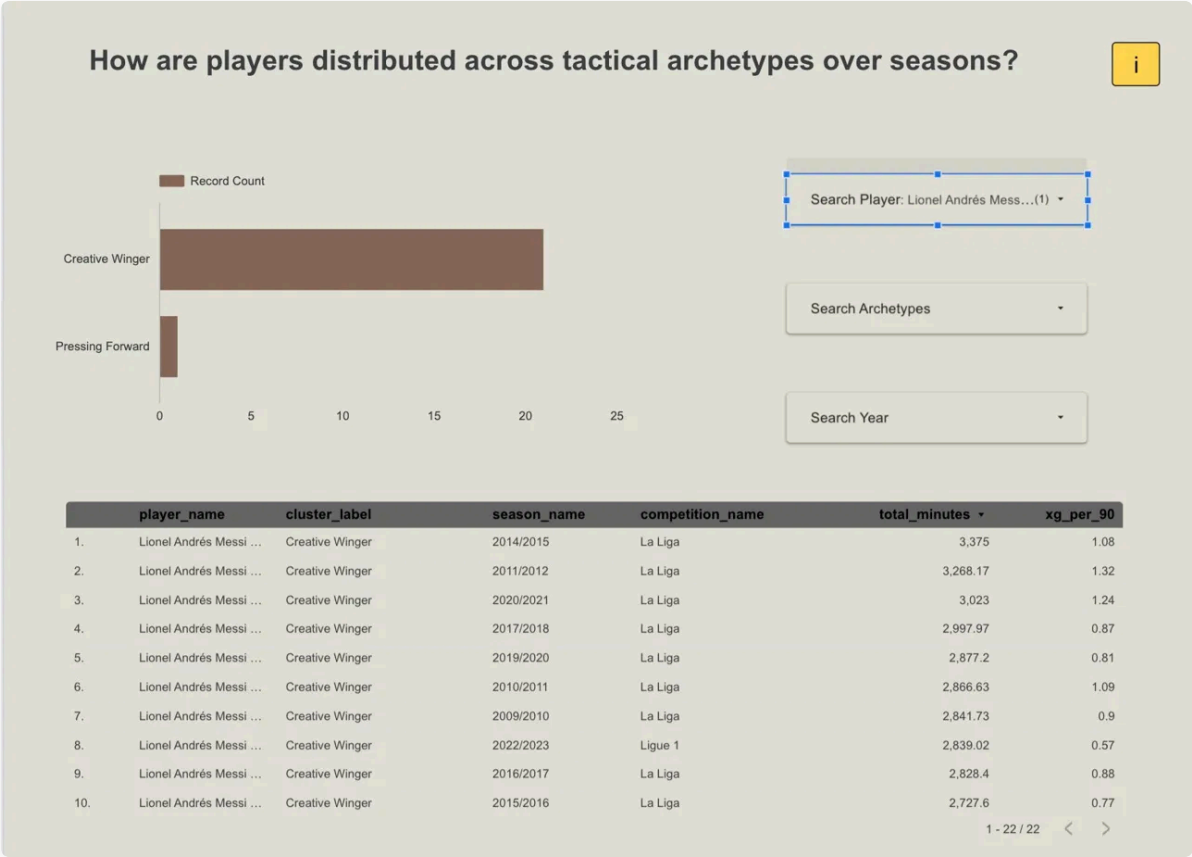
PC1 = overall activity level (left = low, right = high) · PC2 = defensive vs attacking profile (top = defensive) · colour = archetype

Player Archetypes Dashboard — Overview



5,759 player-seasons · 15 competitions · Low Activity and Defensive Anchor are the largest groups · filter by archetype, year, or competition

Player Archetypes Dashboard — Explorer



Messi: Creative Winger across 22 La Liga seasons

What Trilemma can do with this:

- **Scouting** — filter by archetype to find undervalued players in lower-budget leagues
- **Squad balance** — check archetype mix before a transfer window
- **Player tracking** — monitor how a player's style evolves across seasons

Consistency: The Question & Pipeline

RQ2: Do players perform differently for club vs national team?

```
int_player_club_vs_national → consistency.py → consistency_scores → Looker Explorer
```

- **Eligibility:** ≥ 270 minutes in **both** contexts (~3 full matches each)
- **Cohort:** 493 dual-context players across **58 countries**
- **Method:** Z-scores computed **within each context** separately — club peers vs club, national vs national

Consistency Formulas & Results

Z-score standardisation (within each context):

$$Z\text{-score} = \frac{x - \mu_{\text{context}}}{\sigma_{\text{context}}}$$

Performance score = Σ (Z-score $\times$ PCA weight) — club & national axes on the scatter

Consistency score = $1 - \text{mean}(|Z_{\text{club}} - Z_{\text{national}}|)$ — profile similarity, not quality level

Results: 493 players · 58 countries · ~25% per quadrant · **Germany** leads national score · **Sweden** lowest in cohort

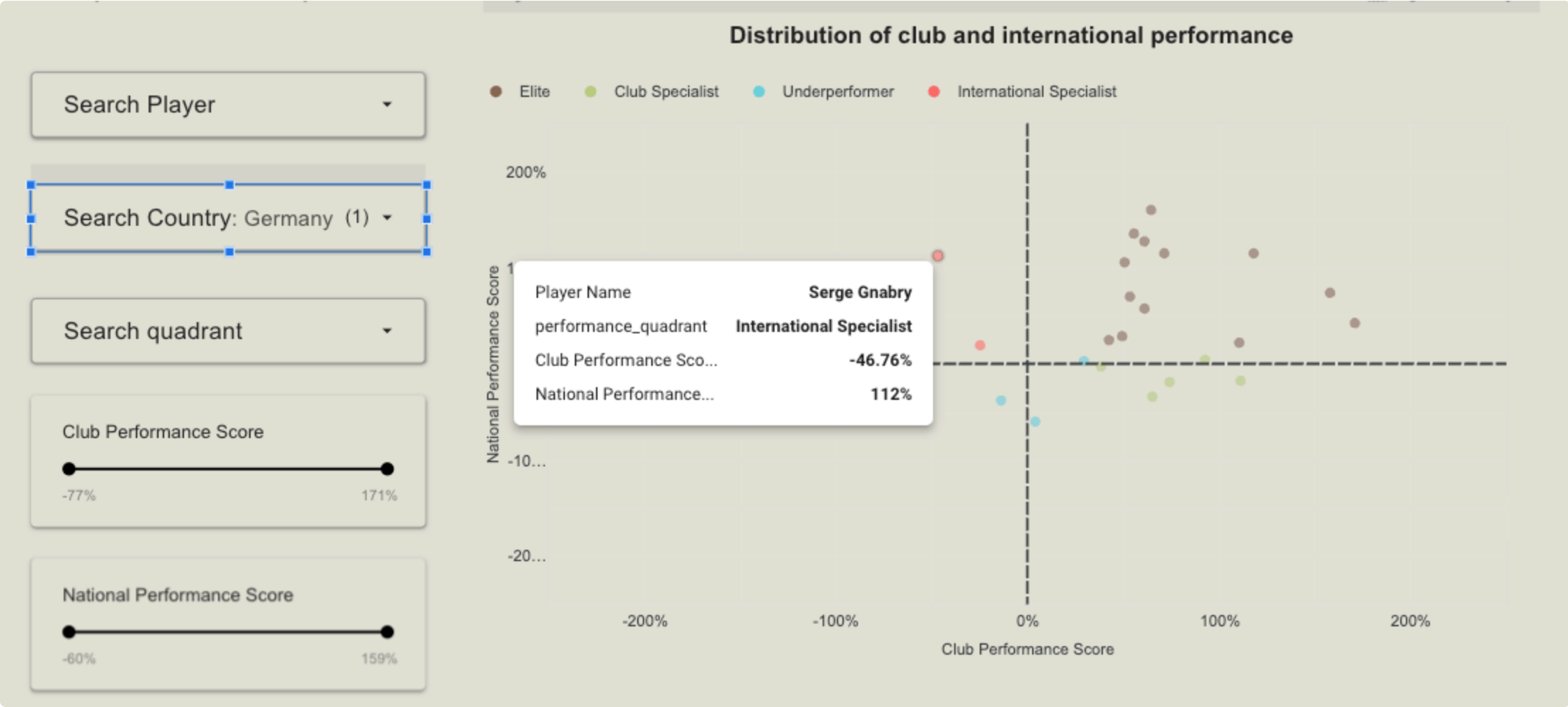
Quadrant Definitions

Split cohort at **medians** of club and national performance score (~25% each)

Quadrant	Definition
Elite	$\geq$ median performance in both contexts
Club Specialist	Club $\geq$ median · national $<$ median
International Specialist	Club $<$ median · national $\geq$ median
Underperformer	Below median in both contexts

Consistency Explorer — Quadrant View

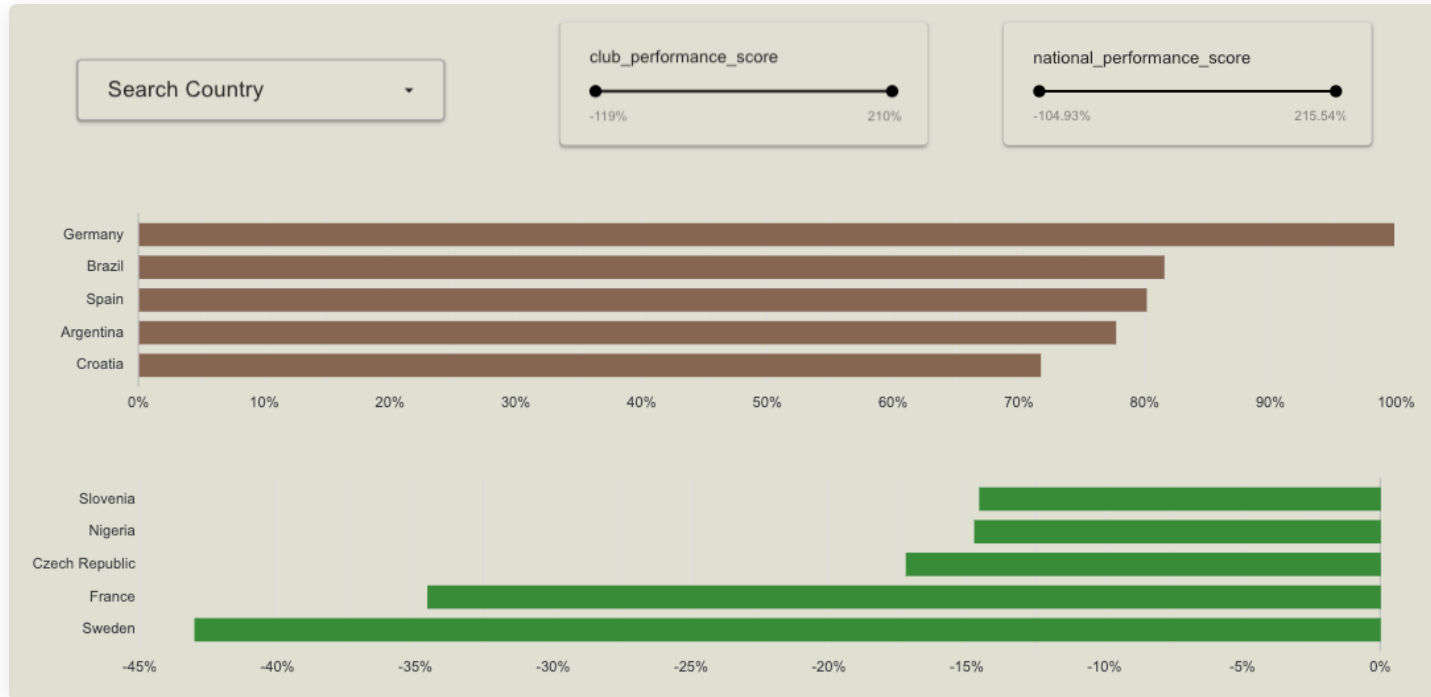
RQ2: Club score (x) vs national score (y) — each point = one player



X-axis = club performance score · Y-axis = national performance score · quadrants split at medians · 493 players

Consistency Explorer — Country Rankings

RQ2: National performance score by country



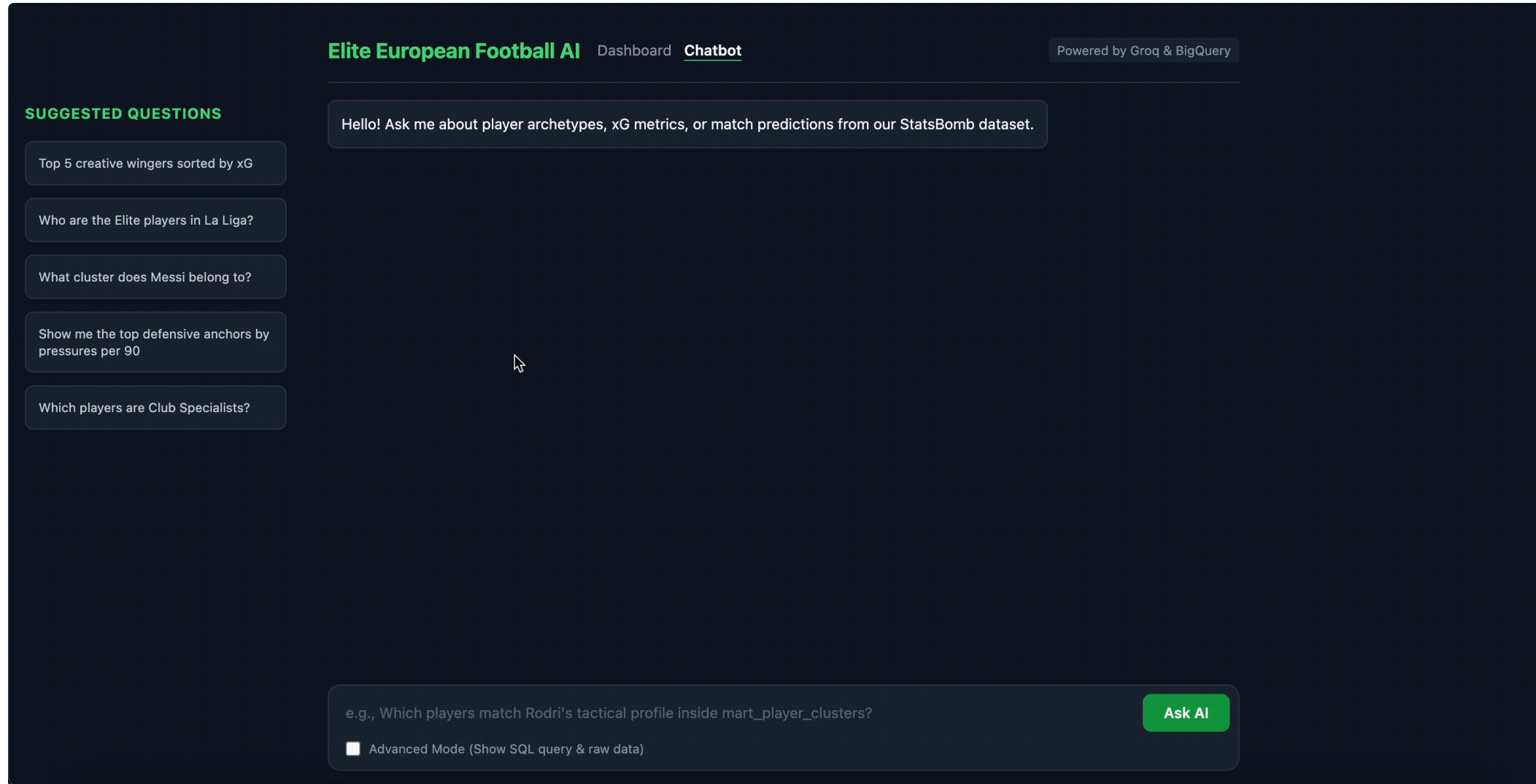
Germany leads · Sweden lowest · bar chart scaled to full range

What Trilemma can do with this:

- **National selectors** — target International Specialists who outperform for country
- **Club recruitment** — Club Specialists may be undervalued on the transfer market
- **Next step** — extend to women's football; broaden international competition coverage

LLM chatbot

RQ3: Can analysts query football data in plain English?

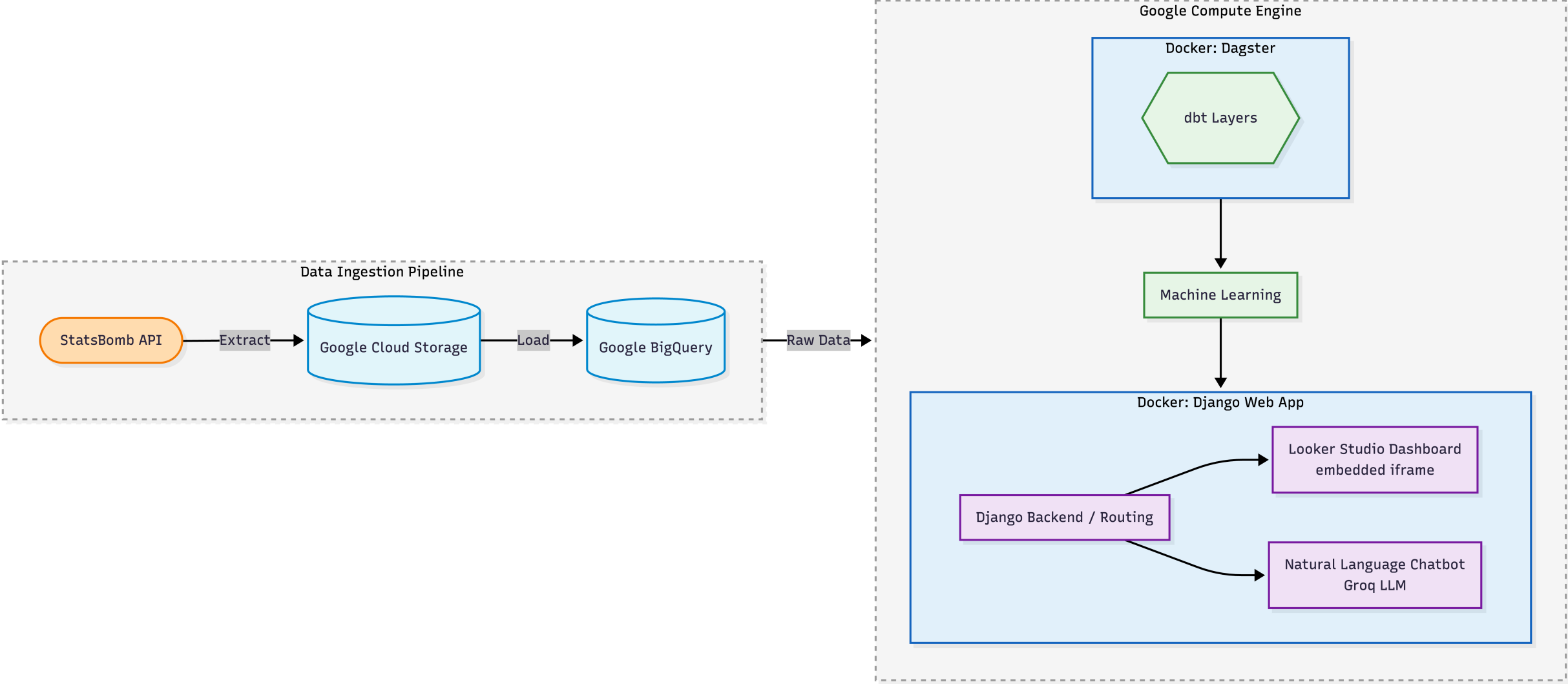


Plain-English question → BigQuery SQL → formatted answer

LLM chatbot — How It Works

- **Engine:** Groq `llama-3.3-70b-versatile`
- **Tool-calling:** English → BigQuery SQL → formatted answer
- **Advanced mode:** shows generated SQL for debugging

System Architecture — End to End



StatsBomb API → GCS → BigQuery → Dagster (dbt + ML) → Django (GCE) → Looker Studio · Chatbot

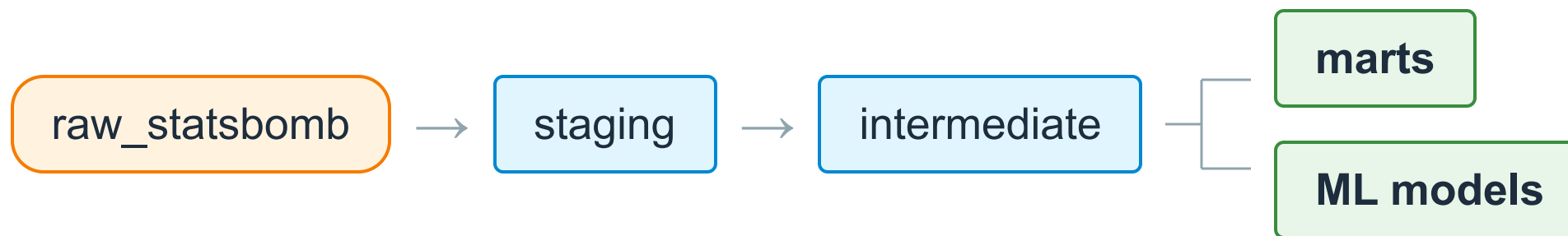
Architecture — Components

- **Ingestion:** StatsBomb API → Google Cloud Storage (GCS)
- **Warehouse:** Centralized in Google BigQuery
- **Transformation:** 3-layer dbt architecture
- **Machine Learning:** Python scripts for PCA/KMeans and Consistency Scoring
- **Serving:** Django web app · Looker Studio dashboard · LLM chatbot

Data Transformation — dbt

dbt (data build tool) = **version-controlled SQL with automated testing** — like GitHub Actions for your data warehouse: define each transform as a SQL SELECT, declare dependencies, dbt runs them in order and tests each output.

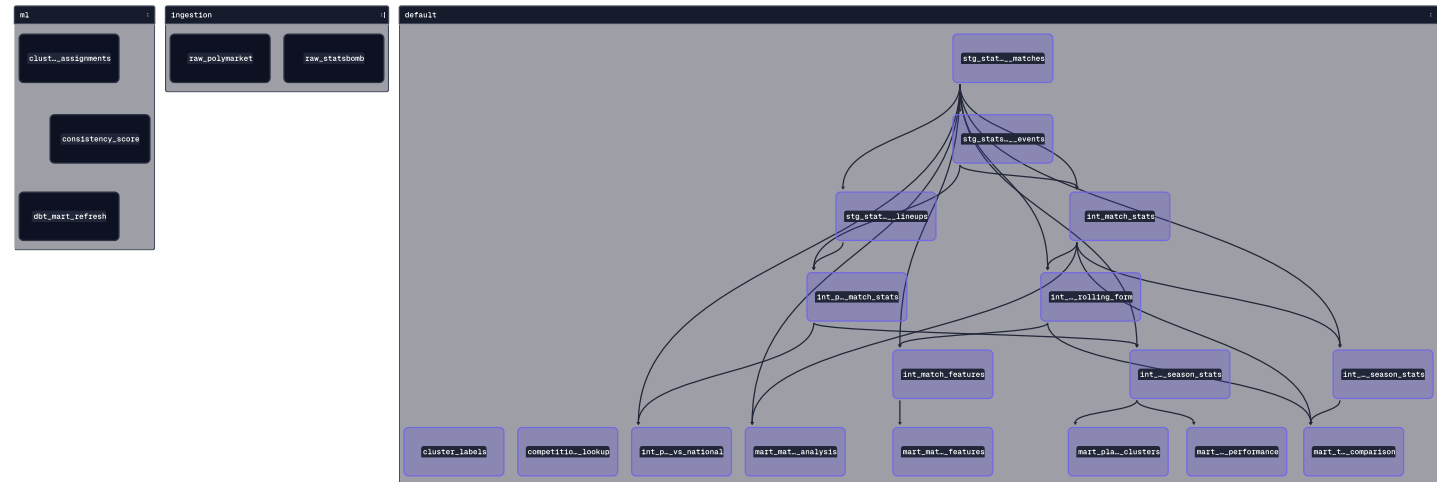
- **Staging** — 3 models · type casts only · 1:1 mirror of raw StatsBomb tables
- **Intermediate** — 7 models · 270-min filter · compute `x_*` per-90 metrics · men's competitions only
- **Marts** — 5 models · join ML outputs (cluster assignments + consistency scores) · final tables for Looker + chatbot



Pipeline Orchestration — Dagster

Dagster = **automated scheduler + dependency graph** — like GitHub Actions job order for data: knows what must run before what, and surfaces failures visually.

- **Weekly schedule** — full pipeline every Monday 6:00 AM UTC
- **Hourly sensor** — new GCS file → dbt + ML only (skips re-ingestion)
- **Zero manual runs** — fully hands-off for Trilemma after handover
- **Asset graph** — every step tracked; failures surface immediately with clear error messages



Asset graph · ingestion → dbt → ML → marts

Limitations & Future Scope

- **Goalkeeper Filter** — excluded from PCA clustering (`cluster = -1`, archetype = “Goalkeeper”)
- **Data Asymmetry** — open-source StatsBomb data is inherently unbalanced; match volume varies by competition and season
- **Women’s football** — intermediate layer filters to men’s competitions; extending requires parallel dbt models
- **Chatbot scope** — single-step queries only; complex multi-step reasoning not yet supported

Partner Handover — What Trilemma Gets

- Fully documented **GitHub repository**
- **CI/CD Pipeline** → automated tests + automated deployment
- **Quarto** final report with methodology documentation
- `HANDOVER.md` — component-level runbooks for each service
- `docker compose up` → spins up Django app + Dagster UI locally

Partner Handover

Partner Handover Guide

UBC MDS Soccer Capstone 2026 — Trilemma Foundation Football Analytics

Last updated: June 2026 **Outgoing team:** Quan Hoang, Li Pu, Teem Kwong, Rabin Duran **For questions contact:** [quandothoang \(GitHub\)](#)

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Thank You

Questions?

Quan Hoang · Li Pu · Teem Kwong · Rabin Duran

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